

Autodesk® Scaleform®

Scaleform 最佳優化指南

Scaleform 4.0

Matthew Doyle

3.0

2011 4 25

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Autodesk® Scaleform® 4.3

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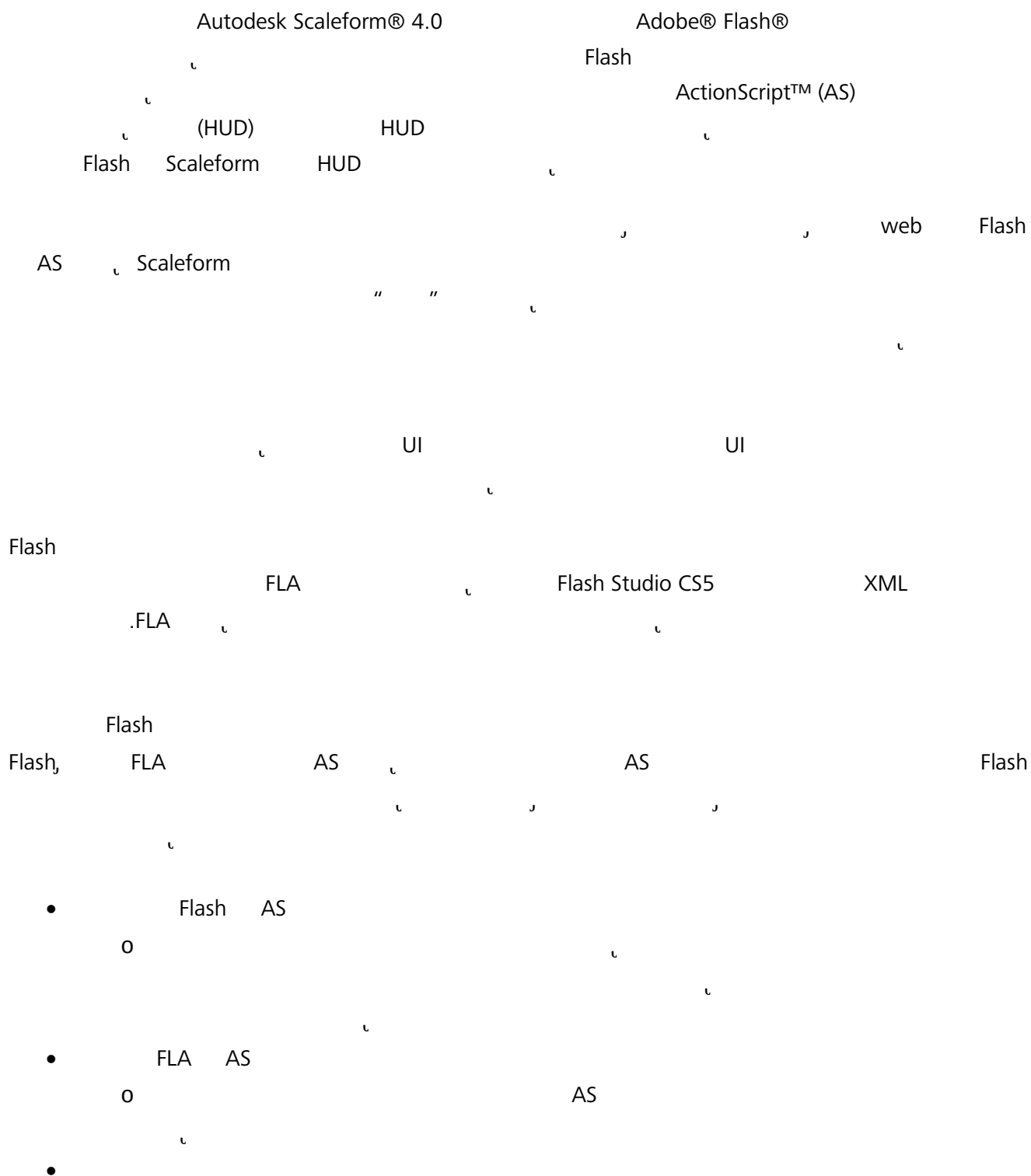
(301) 446-3200

(301) 446-3199

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1 概要



0

-

Flash AS

0

Flash

Flash

Flash AS

Flash

-

-

ActionScript

Flash

Flash

1.1

-

-

UI

Flash

Scaleform 3.3

Scaleform

(CLIK™)

1.2

-

-

Flash

HUD

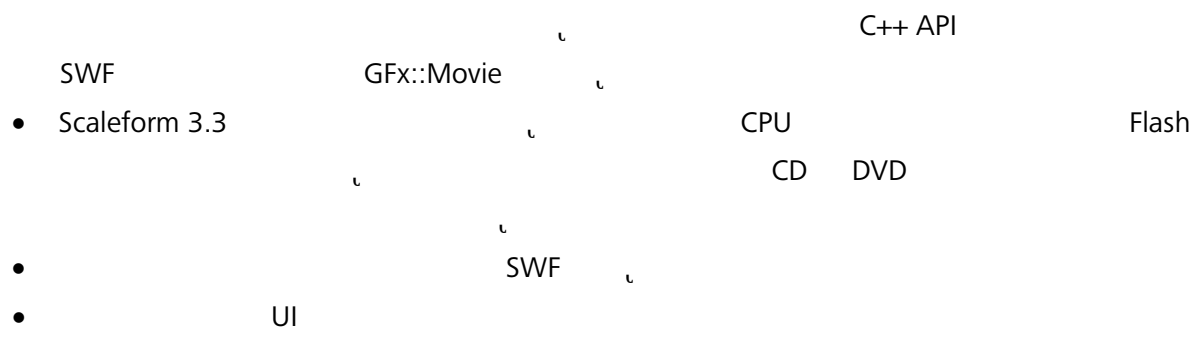
Flash

Flash

MMORPG

在 AS 端，不同文件可以通过 AS2 和

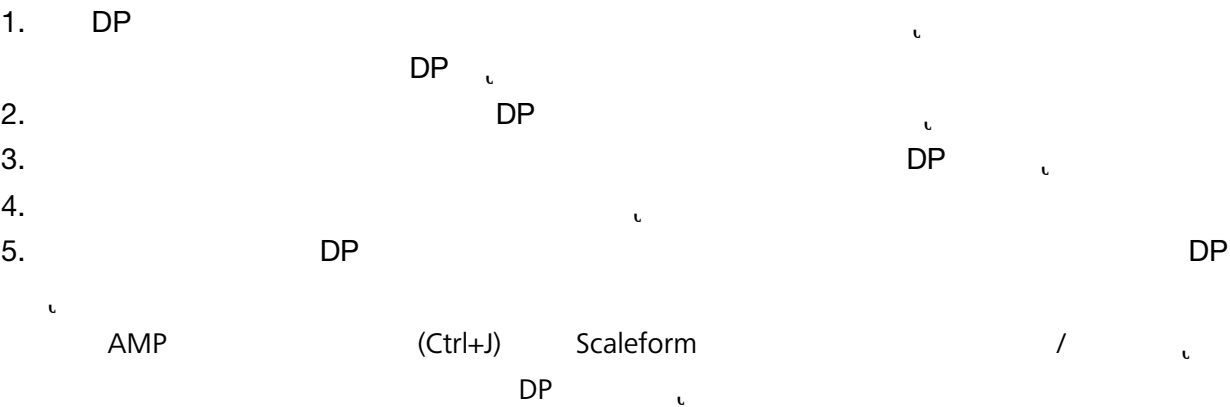
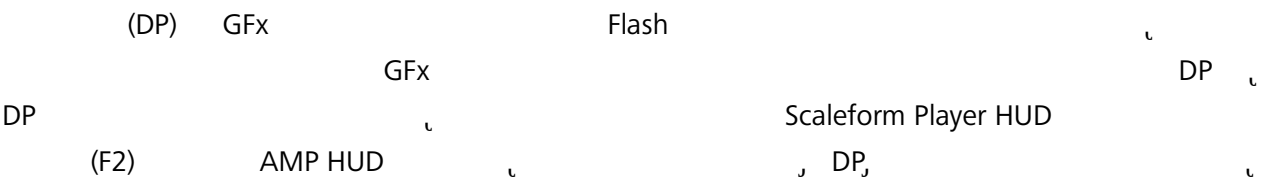
AS3 中的各种 SWF 加载 API 进行加载。



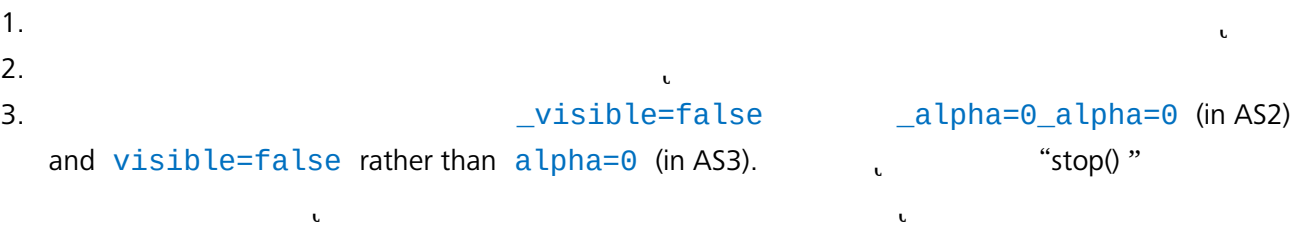
2

Flash

2.1



2.2



2.3

2.3.1

Flash

Scaleform

2.3.2

-
-

0
0

alpha Flash

`_visible`(AS2) or `visible` (AS3) `false`
`_alpha_alpha` (AS2) or `alpha` (AS3) 0, `_alpha alpha` property
100

Flash

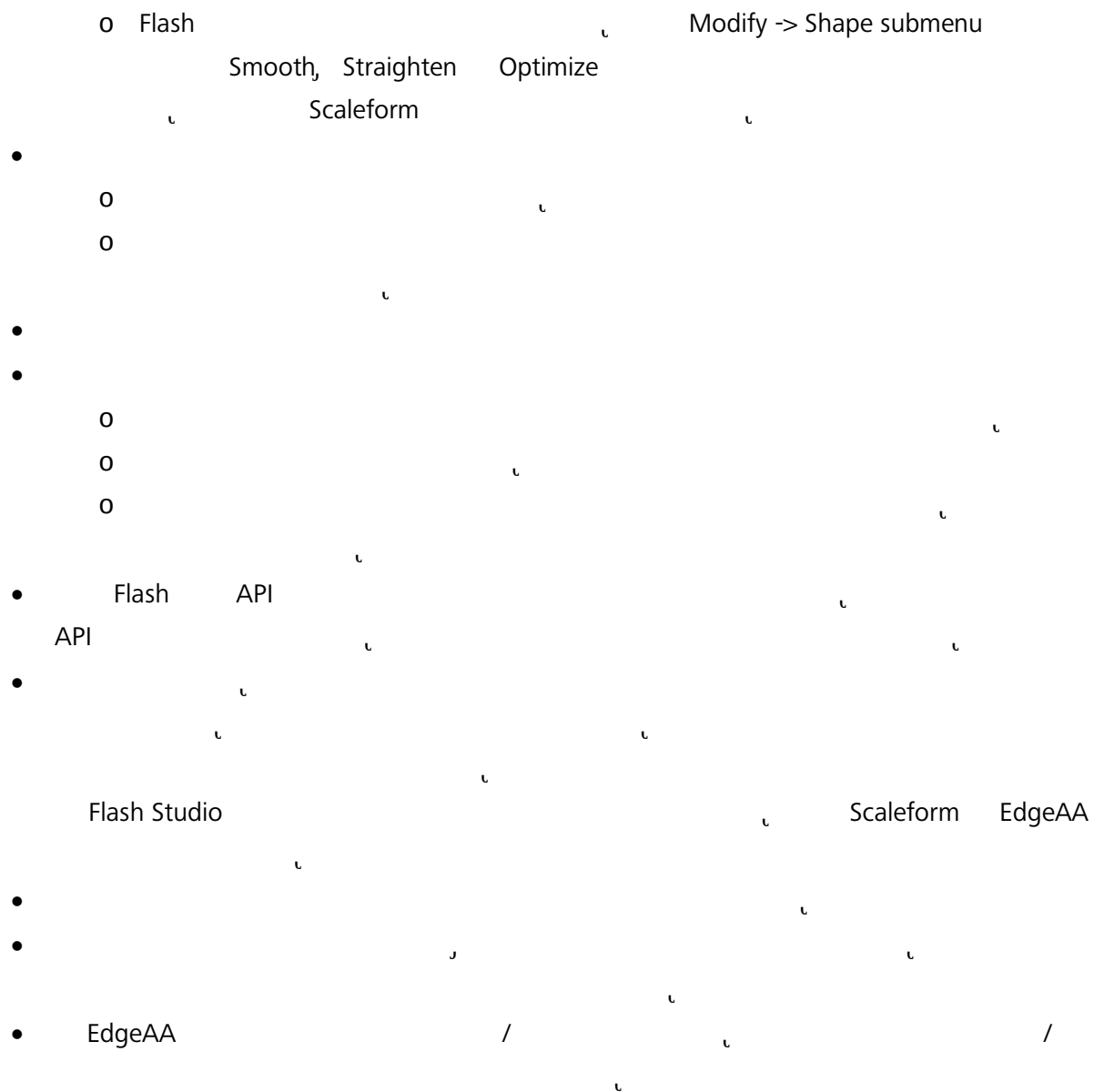
`_x` `_y`(in AS2)

or x and y properties (in AS3)

`_visible /visible` `false` Flash

-

0
0



2.3.3

Flash

SWF _visible false _alpha 0

Scaleform _alpha 0 false CPU ActionScript CPU

SWF false _visible

- / 2 16x32, 256x128,
- 1024x1024 512x32,
- JPEG JPEG
- DXT gfxexport SWF GFX
- DXT DDS gfxexport
- -qp -qh
- /
- Edge AA
- ActionScript
- SWF/GFX SWF 1024 x 1024 JPEG
- 4 MB
- UI gfxexport -i DDS 4
- AMP
- CPU
- gradient.swf SWF gfxexport
- -d0 gradients.swf SWF
- Adobe Photoshop® Flash
- PNG

-
-

Flash

2.3.4

FLA

SWF

AS

12 FPS 24 FPS 12 FPS 24 FPS 12 FPS 24 FPS 5

[onEnterFrame](#) [onEnterFrame](#) (in AS2) or [Event.ENTER_FRAME](#) (in AS3)

[onEnterFrame/Event.ENTER_FRAME](#)

[setInterval](#)

[onEnterFrame/Event.ENTER_FRAME](#)

[setInterval](#)

30 40 FPS
Flash UI

CPU

-
-

on/off

[_visible/visible = true/false](#)

/

-

tween

0

tween

tween

/

0

tween

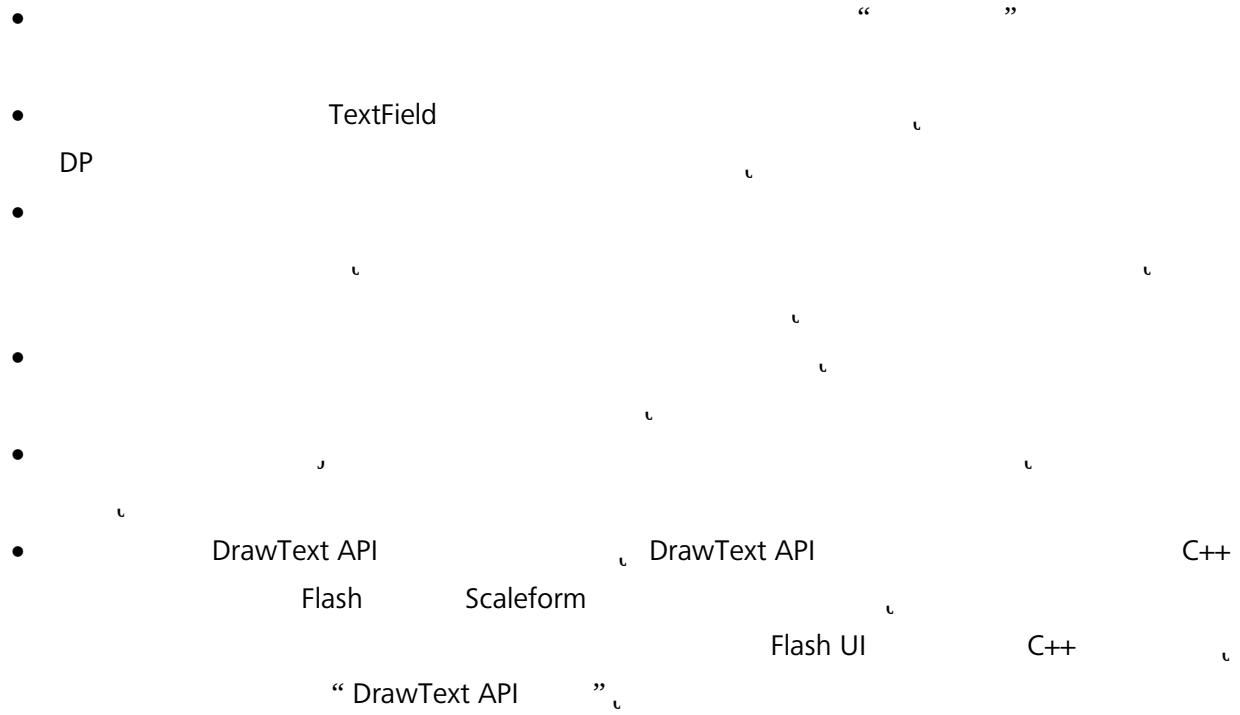
Flash Tween

J J

•

•

•



3 ActionScript

ActionScript



3.1 *ActionScript2*

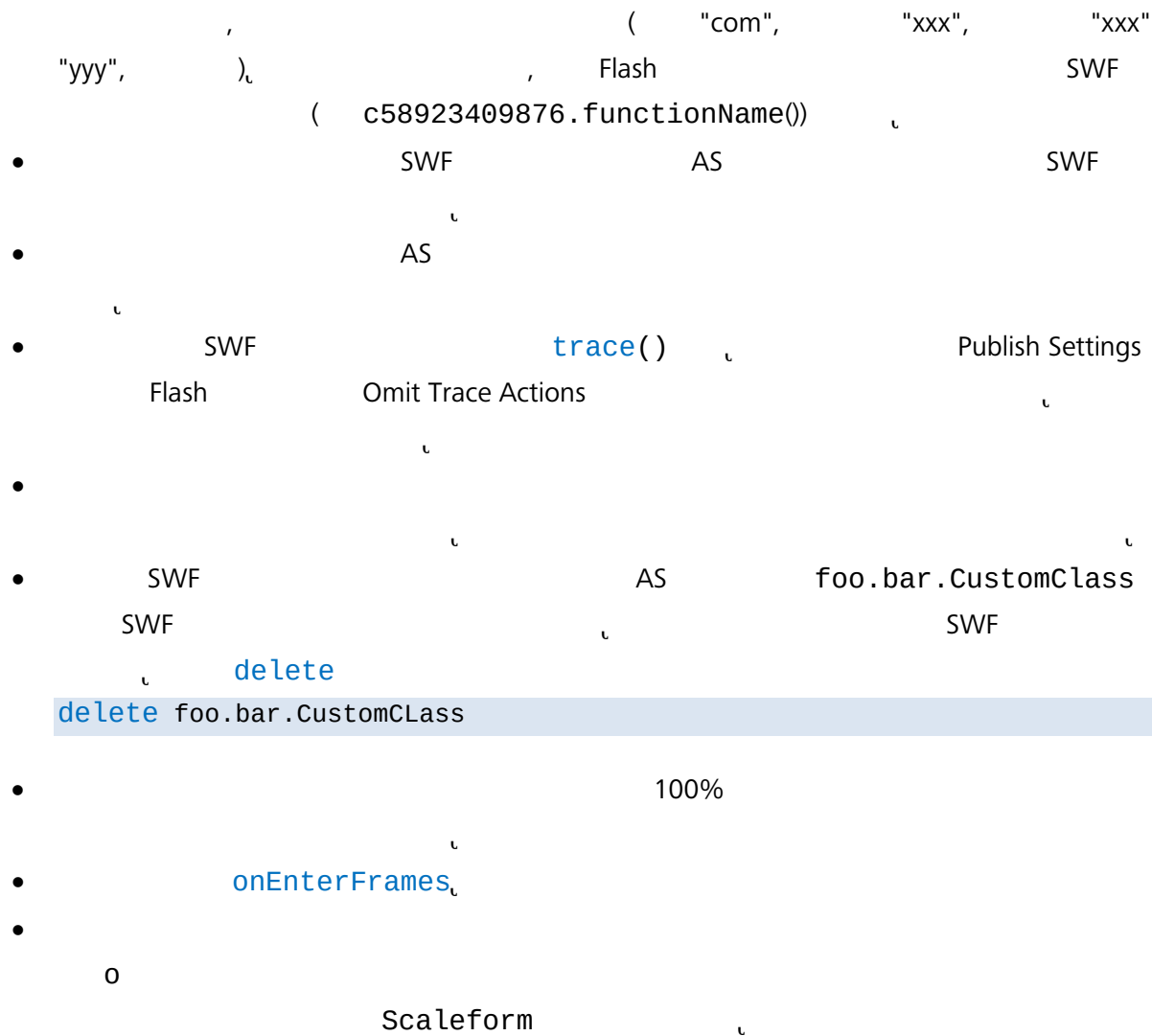
- AS
 - attachMovie FLA
- AS
 - on() onClipEvent() onEnterFrame, onPress
 - “if”
 - Flash AS
 - (onClipEvent, on)
 - (gotoAndPlay, play, stop)
- 避免在包含多個動畫的長時間軸電影剪接中使用“遠距”gotoAndPlay/gotoAndStop。
 - gotoAndPlay/gotoAndStop,
 - ,gotoAndPlay/gotoAndStop
 - gotoAndPlay/gotoAndStop

- 0 gotoAndPlay/gotoAndStop, ,
gotoAndPlay/gotoAndStop
- gotoAndPlay/gotoAndStop
gotoAndPlay/gotoAndStop ,
- gotoAndPlay/gotoAndStop
- C++
[Gfx::Movie::SetVariableArray](#) C++ AS
- Scaleform
Scaleform
- 0 bug
- eval()
- Array.length myArr.length

```
var fontArr:Array = TextField.getFontList();
var arrayLen:Number = fontArr.length;
for (var i:Number = 0; i < arrayLen; i++) {
    trace(fontArr[i]);
}
```

```
var fontArr:Array = TextField.getFontList();
for (var i:Number = 0; i < fontArr.length; i++) {
    trace(fontArr[i]);
}
```

- null
- removeListener()
- AS VM
AS "if"
"com.xxx.yyy.aaa.bbb",VM "com", "xxx", "yyy", "aaa", "bbb",
"if" / /



3.1.1



3.1.2

-
- `with`

3.1.3 /

-
- “ var ” “ var ” ActionScript
-
-
-
- `null` SWF `null`
-
- AS

3.2 一般 *ActionScript 3* 指导原则

AS3

3.2.1 严格数据类型确定

-
- (Object) AS3 “
-
- (Array) (Vector) 4294967295 /
- Vector<*>
- Vector<*>
- Vector<int>, Vector<String> Vector<YourFavoriteClass>
- Vector<*>
- flash.utils.Dictionary
- Gfx Flash
- / AS2 AS3
- AS3

3.3

1. AS `onEnterFrame/ Event.ENTER_FRAME`
2. UI
3. `(_visible(AS2) or visible (AS3) true`
`stop()` Advance
4. `_global.noInvisibleAdvance(AS2) or`
`scaleform.gfx.Extensions.noInvisibleAdvance (AS3)`
Advance “true”
- Flash Flash
`_global.gfxExtensions true` Scaleform
- 5.
- 6.

3.4 *onEnterFrame and Event.ENTER_FRAME*

- `onEnterFrame(AS2) or Event.ENTER_FRAME (AS3)`
- `onEnterFrame/Event.ENTER_FRAME`
- `setInterval setTimeout setInterval`
- `clearInterval`
- `setInterval setTimeout onEnterFrame/Event.ENTER_FRAME`
`onEnterFrame/Event.ENTER_FRAME`

3.4.1 清理 `onEnterFrame`

`delete onEnterFrame`

```
delete this.onEnterFrame;
delete mc.onEnterFrame;
```

```
    null    undefined    onEnterFrame(    this.onEnterFrame = null;)
    onEnterFrame          ↳ Scaleform
onEnterFrame          ↳
```

3.5 ActionScript 2 Optimizations

3.5.1 onClipEvent 和 on Events

```
onClipEvent()    on()          onEnterFrame,    onPress    ↳
```

- -
- ```
 onClipEvent
this, _global, arguments, super ↳ 256
```

```
 onLoad
onClipEvent(construct) onEnterFrame
```

```
onClipEvent(construct)
{
 this.onLoad = function()
 {
 //
 }
}
```

```
onClipEvent(load) ↳
```

### 3.5.2 Var 關鍵字

```
var ↳ AS
var ↳ AS ↳
```

```
var i = 1000;
countIt = function()
{
 num = 0;
```

```

 for(j=0; j<i; j++)
 {
 j++;
 num += Math.random();
 }
 displayNumber.text = num;
}

```

```

var i = 1000;
countIt = function()
{
 var num = 0;
 var ii = i;
 for(var j=0; j<ii; j++)
 {
 j++;
 num += Math.random();
 }
 displayNumber.text = num;
}

```

### 3.5.3 預存

var

```

function foo(var obj:Object)
{
 for (var i = 0; i < obj.size; i++)
 {
 obj.value[i] = obj.num1 * obj.num2;
 }
}

```

```

function foo(var obj:Object)
{
 var sz = obj.size;
 var n1 = obj.num1;
 var n2 = obj.num1;
}

```

```

 for (var i = 0; i < sz; i++)
 {
 obj.value[i] = n1*n2;
 }
}

```

```

var floor = Math.floor
var ceil = Math.ceil
num = floor(x) - ceil(y);

```

```

var keyDown = Key.isDown;
var keyLeft = Key.LEFT;
if (keyDown(keyLeft))
{
 //
}

```

### 3.5.4 預存長路徑

```

mc.ch1.hc3.djf3.jd9._x = 233;
mc.ch1.hc3.djf3._x = 455;

```

```

var djf3 = mc.ch1.hc3.djf3;
djf3._x = 455;

var jd9 = djf3.jd9;
jd9._x = 223;

```

### 3.5.5 複雜運算式

```

, C
this[_global.mynames[queue]][_global.slots[i]].gosplash.text =
_global.MyStrings[queue];

```

```

var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
var slot_i = _global.slots[i];
_splqueue[slot_i].gosplash.text = _splstring;

```

```

for(i=0; i<3; i++)
{
 this[_global.mynames[queue]][_global.slots[i]].gosplash.text =
 _global.MyStrings[queue];
 this[_global.mynames[queue]][_global.slots[i]].gosplash2.text =
 _global.MyStrings[queue];
 this[_global.mynames[queue]][_global.slots[i]].gosplash2.textColor =
 0x000000;
}

```

```

var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
for (var i=0; i<3; i++)
{
 var slot_i = _global.slots[i];
 _splqueue[slot_i].gosplash.text = _splstring;
 _splqueue[slot_i].gosplash2.text = splstring;
 _splqueue[slot_i].gosplash2.textColor = 0x000000;
}

```

6

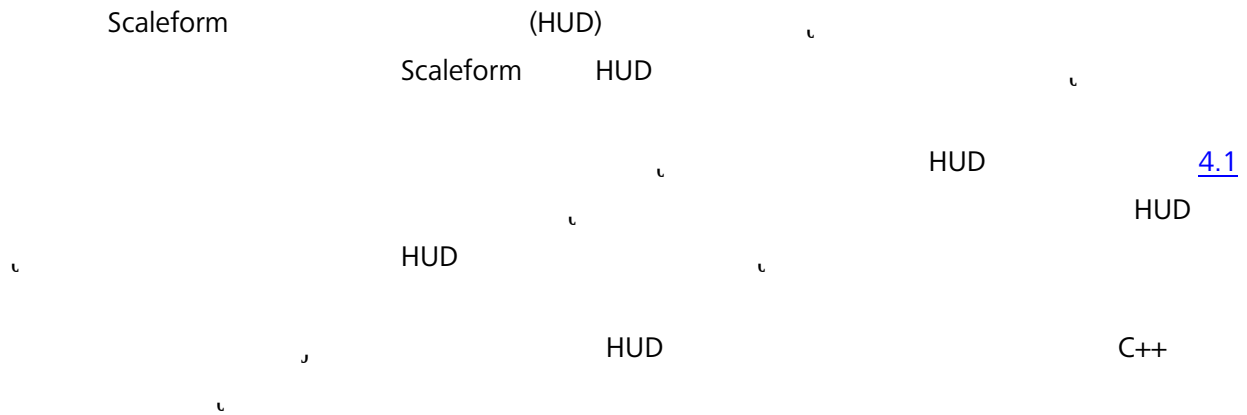
```

var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
for (var i=0; i<3; i++)
{
 var slot_i = _global.slots[i];
 var elem = _splqueue[slot_i];
 elem.gosplash.text = _splstring;
 var gspl2 = elem.gosplash2;
 gspl2.text = splstring;
 gspl2.textColor = 0x000000;
}

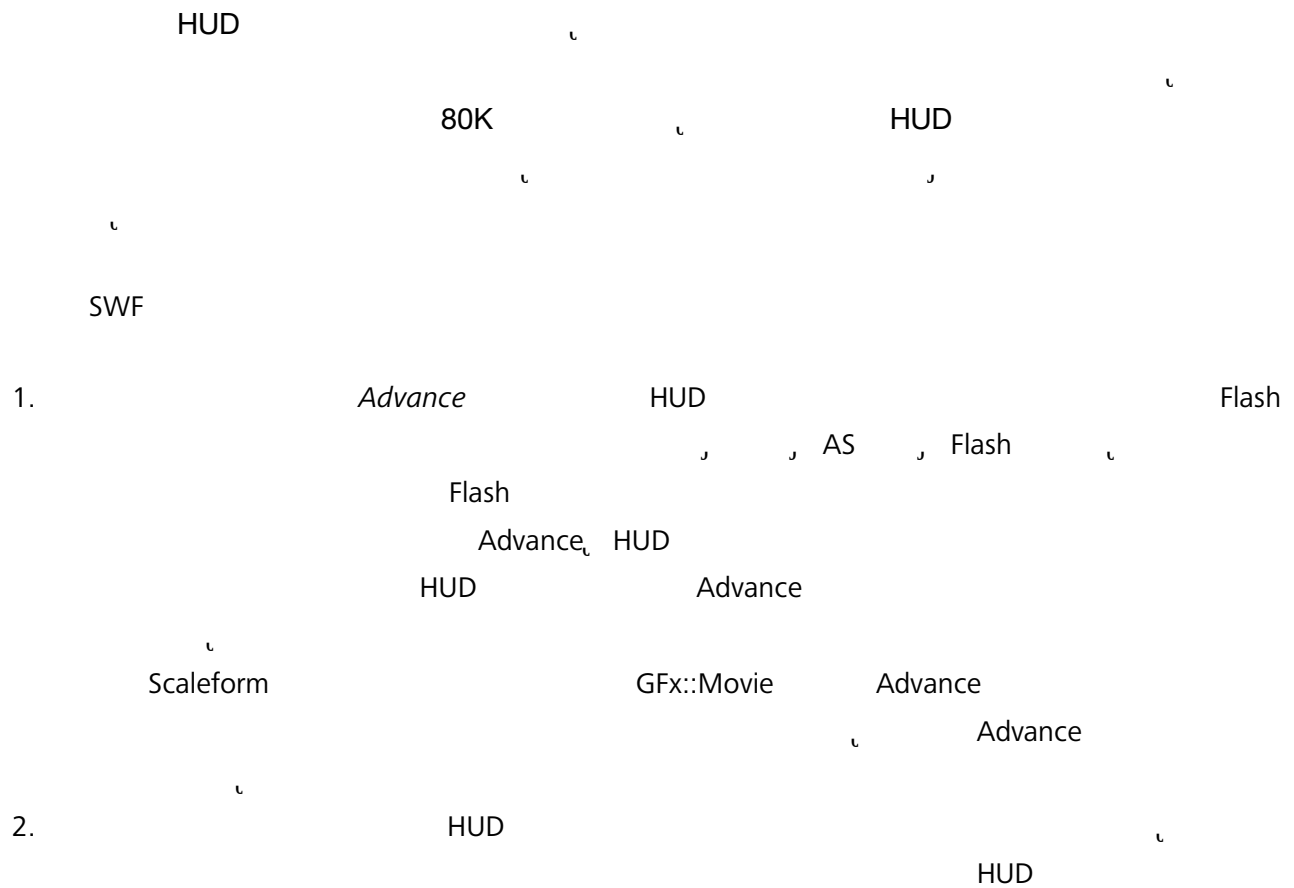
```



## 4 HUD



## 4.1 SWF



## 4.2

## SWF

Flash AS `loadMovie`

Flash

1. `_visible = false` (AS2) or `visible = false` (AS3)

`_global.noInvisibleAdvance` (AS2) or  
`scaleform.gfx.Extensions.noInvisibleAdvance` (AS3) `true`

**HUD**

`_visible/visible = false`

Advance

Flash

HUD

HUD

Advance

`_global.gfxExtensions` (AS2) or `scaleform.gfx.Extensions.enabled` (AS3)

`true` Scaleform

2. `SetVariableArray`

`SetVariableArray`

3. `onEnterFrame` (AS2) or `Event.ENTER_FRAME` (AS3)

HUD `onEnterFrame//Event.ENTER_FRAME`

Advance

4. AS

Flash

30-40 FPS

15-20 FPS

HUD

6

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U

## 4.2

Flash

Flash

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C++

RenderString

Flash

AS

***(Advanced)***

U

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HUD

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HUD

Advance<sub>t</sub>

CPU

Render::Renderer<sub>l</sub>

C++

Render::Renderer

HUD

Render::Renderer

Advance

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2.

U

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23

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2.1

" \_\_\_\_\_ "

u

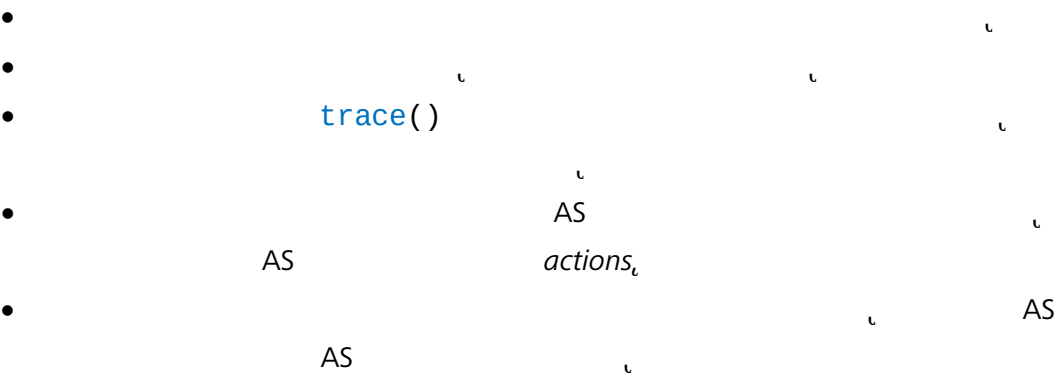
FAQ<sub>u</sub>

5

5.1 Flash



5.2



## 6

ActionScript 2.0

[http://www.adobe.com/devnet/flash/articles/as\\_bestpractices.html](http://www.adobe.com/devnet/flash/articles/as_bestpractices.html)

Flash 8

[http://www.adobe.com/devnet/flash/articles/flash8\\_bestpractices.html](http://www.adobe.com/devnet/flash/articles/flash8_bestpractices.html)

Flash ActionScript 2.0

[http://www.adobe.com/devnet/flash/articles/actionscript\\_guide.html](http://www.adobe.com/devnet/flash/articles/actionscript_guide.html)